

First Pair Loop Lab

A Kiffness-derived proof book for human-AI publishing

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First Pair Press

Start Here

This proof book is small on purpose.

The source idea comes from the Kiffness controller-manual workflow: teach a musician how to turn a compact MIDI controller into an actual practice loop. The publishing idea comes from First Pair Press: make the book itself a visible collaboration between a human and an AI.

The human brings intention:

- this should feel practical, not ceremonial;
- the reader should know what to do with their hands;
- the source should remain readable years from now;
- the pipeline should explain itself.

The AI brings friction removal:

- convert source into renderer formats;
- check that artifacts were produced;
- preserve exact warnings;
- keep the workflow legible for the next collaborator.

That is the first pair.

The Rig

The musical rig is deliberately modest:

- an Akai MPK mini or similar small controller;
- GarageBand or Ableton Live;
- one drum sound;
- one bass sound;
- one chord sound;
- one lead or sample sound.

The publishing rig is equally modest:

- First Pair troff source for the Bell Labs path;
- Markdown for the portable semantic path;
- Pandoc for structured conversion;
- Typst for the modern PDF;
- EPUB for readers;
- Neatroff and utmac for the classic page.

Neither rig tries to impress by size. The point is repeatability.

A First Loop

Set the tempo to something slow enough to hear your mistakes. Ninety-two beats per minute is a good beginner speed.

Make four tracks.

1. Drums.
2. Bass.
3. Chords.
4. Lead or sample.

Record one bar of drums first. Use the pads if the controller has pads. Use the keyboard if it does not. The pattern can be simple: kick on one and three, snare on two and four, closed hat on every eighth note.

Now record one bar of bass. Use only the root and fifth. If the song is in A minor, that means A and E. Do not chase cleverness. Make the loop stable.

Add two-note chords. A minor can be A and C. F can be F and A. C can be C and E. G can be G and B. Two notes are enough to teach the hand what harmony feels like.

Finally add a lead sound or a sample. Give it space. One short answer every two bars is better than a flood of notes.

What The Controller Teaches

A small controller is useful because it refuses to let you hide.

The keys teach pitch. The pads teach rhythm. The knobs teach motion. The octave buttons teach range. The transport buttons teach the difference between recording and performing.

The beginner mistake is to map everything at once. The better workflow is to assign one job at a time:

- keys play notes;
- pads play drums or launch clips;
- knobs shape one filter and one reverb;
- transport starts, stops, and records;
- the computer screen confirms what happened.

When the mapping is boring, the practice becomes musical.

What The Publishing Pipeline Teaches

The book pipeline follows the same rule.

First Pair troff is the studio patch sheet: direct, plain, and explicit. Markdown is the portable sketch-book. Pandoc is the hub. Typst is the polished software instrument. Neatroff is the classic instrument: terse, programmable, and close to the Bell Labs tradition.

utmac is the macro layer: it gives troff a more book-like vocabulary, with headings, metadata, summaries, links, notes, and typography decisions living in plain text.

Source Contract

The final PDF does not need to show every workshop note. The repository does. AI notes, generated files, warning logs, and manifests are part of the source contract because they tell the next collaborator what actually happened.

The result is not one perfect renderer. The result is a system where renderers can disagree without threatening the source.

How A Human And AI Should Work

The human should write what matters.

The AI should make the workshop easier to use.

A good AI collaborator can draft a table of contents, convert a chapter into utmac, notice a missing artifact, or explain why a PDF has warning output. A bad AI collaborator hides uncertainty and produces a polished artifact whose source nobody can inspect.

First Pair Press chooses the first kind.

For every build, leave evidence:

- the Markdown source;
- the pure First Pair troff source;
- the generated ms source;
- the generated utmac source;
- the Typst PDF;
- the Neatroff PDF;
- the EPUB;
- the VERSION.md manifest;
- the warning log when an engine complains.

That evidence is not bureaucracy. It is trust.

The Practice Page

Use this page as a rehearsal.

Read the source. Build the book. Open the PDF. Open the EPUB. Inspect the First Pair troff source, then inspect the generated utmac source. Read the log. Then change one sentence and build again.

If the sentence changes everywhere, the pipeline works.

If the source remains clear, the human can keep writing.

If the generated files explain what happened, the AI can keep helping.

That is the whole proof.